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- a) 110
b) 001
c) 010
d) 101
259. What is the 1's complement of the binary number 1000101?
a) 1000101
b) 0111010
c) 0111011
d) 1000100
260. What is the 1's complement of the binary number 1110000?
a) 1110000
b) 0001111
c) 0001110
d) 1111111
261. What is the 2's complement of the binary number 11001?
a) 11001
b) 00110
c) 00111
d) 11110
262. What is the 2's complement of the binary number 1010101?
a) 1010101
b) 0101010
c) 0101011
d) 1010100
263. What is the 2's complement of the binary number 111000?
a) 111000
b) 000111
c) 000110
d) 111000
264. What is the 2's complement of the binary number 1001011?
a) 1001011
b) 0110100
c) 0110101
d) 1001010
265. What is the 2's complement of the binary number 11111?
a) 11111
b) 00000
c) 00001
d) 11111
266. What is the 2's complement of the binary number 1011?
a) 1011
b) 0100
c) 0101
d) 1101
267. What is the 2's complement of the binary number 11111111?
a) 11111111
b) 00000000
c) 00000001
d) 11111111
268. What is the 2's complement of the binary number 110?
a) 110
b) 001
c) 010
d) 010

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269. What is the 2's complement of the binary number 1000101?

- a) 1000101
- b) 0111010
- c) 0111011
- d) 0111011

270. What is the 2's complement of the binary number 1110000?

- a) 1110000
- b) 0001111
- c) 0001110
- d) 0001111

271. Subtract 1 from 7 using 1's complement in base 10.

- a) 5
- b) 6
- c) 7
- d) 8

272. When signed numbers are used in binary arithmetic, then which one of the following notations would have unique representation for zero.

- (a) Sign-magnitude.
- (b) 1's complement.
- (c) 2's complement.
- (d) 9's complement

273. Subtract 5 from 9 using 1's complement in base 10.

- a) 3
- b) 4
- c) 10
- d) -4

274. Subtracting 7_{10} from 1_{10} using 1's complement.

- a) 1000
- b) 1100
- c) 1110
- d) 1111

275. Subtracting 8_{10} from 4_{10} using 1's complement.

- a) 1000
- b) 1010
- c) 1100
- d) 1111

276. Subtracting 1_{10} from 7_{10} using 2's complement.

- a) 0101
- b) 0110
- c) 1011
- d) 1101

277. Subtracting 5_{10} from 7_{10} using 2's complement.

- a) 0001
- b) 0100
- c) 0101
- d) 1010

278. Subtracting 2_{10} from 7_{10} using 2's complement.

- a) 0001
- b) 0100
- c) 0101
- d) 0111

279. The 2's complement of the number 1101101 is

- (a) 0101110
- (b) 0111110
- (c) 0110010
- (d) 0010011

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280. -8 is equal to signed binary number

- (a) 10001000
- (b) 00001000
- (c) 10000000
- (d) 11000000

281. Gray code, also known as reflected binary code, is a binary numeral system where two successive values differ by how many bits?

- a) One bit
- b) Two bits
- c) Three bits
- d) Four bits

282. Which of the following statements is true about Gray code?

- a) It is a decimal numbering system.
- b) It is an octal numbering system.
- c) It is a non-weighted numbering system.
- d) It is a self-complementary numbering system.

283. How many bits are required to represent 8 distinct values in Gray code?

- a) 2 bits
- b) 3 bits
- c) 4 bits
- d) 8 bits

284. Which of the following is the Gray code representation of the binary number 1011?

- a) 1101
- b) 1110
- c) 1001
- d) 0110

285. Which application commonly uses Gray code for encoding rotary encoders?

- a) Computer graphics
- b) Data encryption
- c) Error detection
- d) Position sensing

286. Which property of Gray code makes it useful for reducing errors in digital systems?

- a) Orthogonality
- b) Linearity
- c) Complementarity
- d) Monotonicity

287. How many different values can be represented using a 4-bit Gray code?

- a) 4
- b) 8
- c) 16
- d) 32

288. Which sequential Gray code is obtained by toggling only one bit at a time?

- a) Binary
- b) Excess-3
- c) Reflected
- d) Natural

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289. Which of the following is an advantage of using Gray code in digital systems?

- a) Ease of conversion to decimal
- b) Compact representation of numbers
- c) Direct compatibility with analog systems
- d) Support for negative numbers

290. Which coding scheme is commonly used in digital-to-analog converters to prevent glitches during transitions?

- a) Binary code
- b) Excess-3 code
- c) Gray code
- d) BCD code

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EXPLANATION

1. The correct answer is a) Binary
Computers use the binary number system, which uses only two digits (0 and 1), to store and process data. Binary digits, also known as bits, form the foundation of digital computing.

2. The correct answer is a) 0 and 1
The binary number system consists of two digits, 0 and 1. These digits are used to represent numbers and perform arithmetic operations in binary form.

3. The correct answer is b) 19
To convert a binary number to its decimal equivalent, we multiply each digit of the binary number by the corresponding power of 2 and sum the results. In this case, $(1 * 2^4) + (0 * 2^3) + (1 * 2^2) + (0 * 2^1) + (1 * 2^0)$ equals $16 + 0 + 4 + 0 + 1$, which is equal to 19.

4. The correct answer is a) Hexadecimal

-The hexadecimal number system is commonly used to represent color values in computer graphics.

Hexadecimal digits (0-9 and A-F) provide a compact and convenient

way to represent a wide range of colors.

5. The correct answer is c) 4 bits

Each hexadecimal digit represents 4 bits. Since each hexadecimal digit can have 16 possible values (0-9 and A-F), it can be represented by 4 bits ($2^4 = 16$).

6. The correct answer is c) 46

To convert a binary number to its octal equivalent, we group the binary digits into sets of three starting from the rightmost side. In this case, 011 011 0 represents the octal number 46.

7. The correct answer is a) Two's complement

Computers use the two's complement representation to represent negative numbers. This system simplifies arithmetic operations on both positive and negative numbers.

8. The correct answer is b) 163

To convert a hexadecimal number to its decimal equivalent, we multiply each digit of the hexadecimal number by the

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corresponding power of 16 and sum the results. In this case, $(10 * 16^1) + (3 * 16^0)$ equals $160 + 3$, which is equal to 163.

9. The correct answer is d) 65536

A 16-bit binary number can represent $2^{16} = 65536$ unique values. Each bit has two possible states (0 or 1), and with 16 bits, we have a total of 2^{16} combinations.

10. The correct answer is d) 10101

To convert a decimal number to binary, we repeatedly divide the decimal number by 2 and keep track of the remainders. The binary equivalent is obtained by reading the remainders in reverse order. In this case, the division process gives us the remainders: 1, 0, 1, 0, 0. Reading the remainders in reverse order gives us the binary number 10101.

11. The correct answer is c) 10

The decimal number system consists of 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.

12. The correct answer is b) Tens

In the number 536.21, the digit "3" is in the tens place value. The place values increase from right to left as ones, tens, hundreds, thousands, and so on.

13. The correct answer is d) 10000

The successor of a number is the next number in sequence. In the decimal number system, the successor of 999 is 1000. The number 10000 is the next number after 999.

14. The correct answer is a) 999

With 4 digits in the decimal number system, the largest number that can be represented is 999. Once you add another digit, it becomes 1000.

15. The correct answer is d) 2.5

When you add 0.75 and 1.25, the sum is 2.5. In decimal arithmetic, the decimal point aligns, and you add the digits in each place value.

16. The correct answer is c) Thousands

In the number 4,761.23, the digit "7" is in the thousands place value. The place values increase from right to left as ones, tens, hundreds, thousands, and so on.

17. The correct answer is c) 0.6

To find the decimal equivalent of a fraction, divide the numerator by the denominator. In this case, 3 divided by 5 equals 0.6.

18. The correct answer is c) 48

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When you multiply 12.5 by 4, the product is 48. In decimal multiplication, you multiply the digits in each place value.

19. The correct answer is c)
Hundred thousands

In the number 72,566.14, the digit "6" is in the hundred thousands place value. The place values increase from right to left as ones, tens, hundreds, thousands, ten thousands, hundred thousands, and so on.

20. The correct answer is c) 0.625

To find the decimal equivalent of a fraction, divide the numerator by the denominator. In this case, 7 divided by 8 equals 0.875.

21. The correct answer is a) 5

In the number 753.41, the digit "5" is in the ones place value. It represents a value of 5.

22. The correct answer is c)
Hundreds

In the number 4,975.32, the digit "9" is in the hundreds place value. The place values increase from right to left as ones, tens, hundreds, thousands, and so on.

23. The correct answer is b) 8,633

The successor of a number is the next number in sequence. In the

decimal number system, the successor of 8,632 is 8,633.

24. The correct answer is b)
99,999.9

Among the given numbers, 99,999.9 is the largest. It has the highest value due to the combination of digits in the ones, tens, hundreds, and thousands place values.

25. The correct answer is a) 10.75

When you add 3.5 and 7.25, the sum is 10.75. In decimal addition, you add the digits in each place value, including the decimal places.

26. The correct answer is b) Tens

In the number 623.704, the digit "6" is in the tens place value. The place values increase from right to left as ones, tens, hundreds, thousands, and so on.

27. The correct answer is a) 0.25

To find the decimal equivalent of a fraction, divide the numerator by the denominator. In this case, 1 divided by 4 equals 0.25.

28. The correct answer is b) 10.25

When you multiply 5.75 by 2, the product is 10.25. In decimal multiplication, you multiply the

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digits in each place value, including the decimal places.

29. The correct answer is c)
Hundred-thousands

In the number 45,673.218, the digit "3" is in the hundred thousands place value. The place values increase from right to left as ones, tens, hundreds, thousands, ten-thousands, hundred-thousands, and so on.

30. The correct answer is c) 0.428
To find the decimal equivalent of a fraction, divide the numerator by the denominator. In this case, 3 divided by 7 equals approximately 0.428.

31. The correct answer is d) The binary number 1010 is equivalent to the decimal number 10.

32. The correct answer is c) The decimal number 11 is represented in binary as 1011.

33. The correct answer is b) The successor of the binary number 1101 is 1110.

34. The correct answer is c) The largest binary number that can be represented using 4 digits is 1111.

35. The correct answer is c) The sum of the binary numbers 1101 and 1010 is 10111.

36. The correct answer is b) The decimal number 5 is represented in binary as 101.

37. The correct answer is c) The binary number 1001 is equivalent to the decimal number 9.

38. The correct answer is a) The product of the binary numbers 101 and 11 is 1011.

39. The correct answer is c) The decimal number 15 is represented in binary as 1111.

40. The correct answer is c) The binary number 11011 is equivalent to the decimal number 27.

41. The correct answer is b) The binary number 11001 is equivalent to the decimal number 18.

42. The correct answer is a) The decimal number 13 is represented in binary as 1101.

43. The correct answer is c) The successor of the binary number 101011 is 1011001.

44. The correct answer is d) The largest binary number that can be represented using 6 digits is 111111.

45. The correct answer is d) The result of subtracting the binary numbers 10110 and 1101 is 11001.

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46. The correct answer is a) The decimal number 27 is represented in binary as 11011.

47. The correct answer is b) The binary number 1001001 is equivalent to the decimal number 41.

48. The correct answer is b) The result of dividing the binary number 101010 by 10 (in binary) is 1010.

49. The correct answer is a) The decimal number 42 is represented in binary as 101010.

50. The correct answer is c) The binary number 1001111 is equivalent to the decimal number 79.

51. The correct answer is d) The binary number system uses 2 digits, 0 and 1.

52. The correct answer is a) Least Significant Bit (LSB) The rightmost digit in a binary number is called the Least Significant Bit (LSB).

53. The correct answer is d) The binary number 1010 is equivalent to the decimal number 10.

54. The correct answer is a) The binary number 1001 represents the decimal number 9.

55. The correct answer is b) The decimal number 15 is represented in binary as 1111.

56. The correct answer is c) The sum of the binary numbers 1011 and 110 is 1001.

57. The correct answer is d) Subtracting the binary number 1010 from 1111 results in 1110.

58. The correct answer is a) The decimal number 25 is represented in binary as 10001.

59. The correct answer is b) The binary number 110110 is equivalent to the decimal number 38.

60. The correct answer is b) The decimal number 63 is represented in binary as 111111.

61. The correct answer is d) 3 The octal number system uses 8 digits, 0 to 7.

62. The correct answer is a) Least Significant Digit (LSD) The rightmost digit in an octal number is called the Least Significant Digit (LSD).

63. The correct answer is c) 26 The octal number 34 is equivalent to the decimal number 26.

64. The correct answer is c) 17 The octal number 17 represents the decimal number 15.

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65. The correct answer is c) 62 The decimal number 42 is represented in octal as 62.

66. The correct answer is a) 104 The sum of the octal numbers 45 and 47 is 104.

67. The correct answer is c) 22 Subtracting the octal number 55 from 77 results in 22.

68. The correct answer is d) 171 The decimal number 65 is represented in octal as 171.

69. The correct answer is c) 39 The octal number 57 is equivalent to the decimal number 39.

70. The correct answer is a) 135 The decimal number 93 is represented in octal as 135.

71. The correct answer is b) 6 The octal number 14 is equivalent to the decimal number 6.

72. The correct answer is c) The decimal number 25 is represented in octal as 41.

73. The correct answer is c) The successor of the octal number 36 is 41.

74. The correct answer is d) The octal number 31 represents the decimal number 19.

75. The correct answer is d) The sum of the octal numbers 26 and 13 is 61.

76. The correct answer is b) The decimal number 50 is represented in octal as 72.

77. The correct answer is c) The octal number 63 is equivalent to the decimal number 43.

78. The correct answer is b) Subtracting the octal number 45 from 72 results in 23.

79. The correct answer is c) The decimal number 81 is represented in octal as 201.

80. The correct answer is b) The octal number 57 is equivalent to the decimal number 39.

81. The correct answer is d) The hexadecimal number 1F is equivalent to the decimal number 31.

82. The correct answer is b) The decimal number 42 is represented in hexadecimal as 2A.

83. The correct answer is a) The successor of the hexadecimal number 3E is 3F.

84. The correct answer is c) The hexadecimal number 1B represents the decimal number 27.

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85. The correct answer is b) The sum of the hexadecimal numbers 2B and 14 is 39.

86. The correct answer is c) The decimal number 56 is represented in hexadecimal as 38.

87. The correct answer is c) The hexadecimal number 5E is equivalent to the decimal number 94.

88. The correct answer is a) Subtracting the hexadecimal number 2F from 4B results in 18.

89. The correct answer is d) The decimal number 72 is represented in hexadecimal as 5A.

90. The correct answer is b) The hexadecimal number 7F is equivalent to the decimal number 127.

91. The correct answer is a) The decimal number 10 is represented in binary as 1010.

92. The correct answer is d) The binary representation of the decimal number 25 is 10101.

93. The correct answer is a) The decimal number 7 is equivalent to the binary number 111.

94. The correct answer is d) The binary number 1111 represents the decimal number 15.

95. The correct answer is b) The decimal number 42 is represented in binary as 101100.

96. The correct answer is b) The binary equivalent of the decimal number 56 is 111000.

97. The correct answer is a) The binary representation of the decimal number 63 is 111111.

98. The correct answer is b) The decimal number 82 is equivalent to the binary number 1010010.

99. The correct answer is d) The binary number 1000001 represents the decimal number 97.

100. The correct answer is a) The binary representation of the decimal number 120 is 1111000.

101. The correct answer is c) The binary number 1010 is equivalent to the decimal number 10.

102. The correct answer is b) The decimal representation of the binary number 11011 is 19.

103. The correct answer is c) The binary number 111 is equivalent to the decimal number 7.

104. The correct answer is d) The decimal number 15 is represented by the binary number 1111.

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105. The correct answer is b) The binary number 10110 represents the decimal number 22.

106. The correct answer is d) The decimal value of the binary number 110001 is 57.

107. The correct answer is b) The binary number 100111 is equivalent to the decimal number 39.

108. The correct answer is c) The decimal number 42 is represented by the binary number 101010.

109. The correct answer is b) The decimal value of the binary number 1100101 is 97.

110. The correct answer is c) The binary number 111101 represents the decimal number 61.

111. The correct answer is a) The decimal number 10 is represented in octal as 8.

112. The correct answer is b) The octal representation of the decimal number 25 is 31.

113. The correct answer is d) The decimal number 7 is equivalent to the octal number 7.

114. The correct answer is b) The octal number 17 represents the decimal number 15.

115. The correct answer is c) The decimal number 42 is represented in octal as 52.

116. The correct answer is c) The octal equivalent of the decimal number 56 is 70.

117. The correct answer is b) The octal representation of the decimal number 63 is 77.

118. The correct answer is c) The decimal number 82 is equivalent to the octal number 122.

119. The correct answer is d) The octal number 157 represents the decimal number 97.

120. The correct answer is c) The decimal number 120 is represented in octal as 160.

121. The correct answer is c) The octal number 12 is equivalent to the binary number 110.

122. The correct answer is b) The binary representation of the octal number 34 is 10100.

123. The correct answer is c) The octal number 56 is equivalent to the binary number 11100.

124. The correct answer is a) The binary number 111101 represents the octal number 77.

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125. The correct answer is d) The octal number 123 is represented in binary as 1111111.

126. The correct answer is c) The binary equivalent of the octal number 232 is 11001010.

127. The correct answer is a) The octal number 456 is represented in binary as 100110110.

128. The correct answer is a) The binary number 11111111 represents the octal number 777.

129. The correct answer is a) The octal number 625 is equivalent to the binary number 110010101.

130. The correct answer is a) The binary representation of the octal number 347 is 11010111.

131. The correct answer is b) The hexadecimal number 1A is equivalent to the decimal number 18.

132. The correct answer is a) The decimal representation of the hexadecimal number 2F is 37.

133. The correct answer is b) The hexadecimal number 5C is equivalent to the decimal number 92.

134. The correct answer is b) The decimal number 125 represents the hexadecimal number 7D.

135. The correct answer is b) The hexadecimal number A3 represents the decimal number 163.

136. The correct answer is b) The decimal equivalent of the hexadecimal number F0 is 240.

137. The correct answer is b) The hexadecimal number 9E is equivalent to the decimal number 158.

138. The correct answer is c) The decimal number 216 represents the hexadecimal number C8.

139. The correct answer is c) The hexadecimal number E5 is equivalent to the decimal number 233.

140. The correct answer is c) The decimal representation of the hexadecimal number 10F is 271.

141. The correct answer is c) The octal representation of the hexadecimal number 3A is 76.

142. The correct answer is b) The hexadecimal representation of the octal number 27 is 17.

143. The correct answer is d) The octal representation of the hexadecimal number 58 is 230.

144. The correct answer is b) The hexadecimal representation of the octal number 55 is 47.

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145. The correct answer is b) The octal representation of the hexadecimal number AB is 313.

146. The correct answer is a) The hexadecimal representation of the octal number 127 is 4F.

147. The correct answer is d) The octal representation of the hexadecimal number C9 is 451.

148. The correct answer is c) The hexadecimal representation of the octal number 345 is 155.

149. The correct answer is b) The octal representation of the hexadecimal number EF is 247.

150. The correct answer is c) The hexadecimal representation of the octal number 632 is 366.

151. The correct answer is a) The binary number 1101 is equivalent to the hexadecimal number A.

152. The correct answer is a) The binary number 101010 is represented in hexadecimal as 2A.

153. The correct answer is a) The binary number 11110000 is equivalent to the hexadecimal number F0.

154. The correct answer is a) The binary number 10011011 is represented in hexadecimal as 9B.

155. The correct answer is a) The binary number 11001010 is equivalent to the hexadecimal number CA.

156. The correct answer is a) The binary number 10111101 is represented in hexadecimal as BD.

157. The correct answer is a) The binary number 11100011 is equivalent to the hexadecimal number E3.

158. The correct answer is a) The binary number 11010111 is represented in hexadecimal as D7.

159. The correct answer is a) The binary number 10011101 is equivalent to the hexadecimal number 9D.

160. The correct answer is a) The binary number 11111111 is represented in hexadecimal as FF.

161. The correct answer is a) The decimal fraction 0.375 is represented in binary as 0.110.

162. The correct answer is c) The decimal fraction 0.625 is represented in binary as 0.101.

163. The correct answer is b) The decimal fraction 0.875 is represented in binary as 0.110.

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164. The correct answer is b) The decimal fraction 0.25 is represented in binary as 0.010.

165. The correct answer is c) The decimal fraction 0.125 is represented in binary as 0.001.

166. The correct answer is a) The decimal fraction 0.5 is represented in binary as 0.1.

167. The correct answer is a) The decimal fraction 0.0625 is represented in binary as 0.0001.

168. The correct answer is b) The decimal fraction 0.875 is represented in binary as 0.111.

169. The correct answer is b) The decimal fraction 0.875 is represented in binary as 0.110.

170. The correct answer is c) The decimal fraction 0.3 is represented in binary as 0.0011.

171. The correct answer is a) The binary fraction 0.101 is equivalent to the decimal number 0.625.

172. The correct answer is b) The binary fraction 0.011 is equivalent to the decimal number 0.375.

173. The correct answer is b) The binary fraction 0.1101 is equivalent to the decimal number 0.65625.

174. The correct answer is a) The binary fraction 0.0011 is equivalent to the decimal number 0.0625.

175. The correct answer is b) The binary fraction 0.10001 is equivalent to the decimal number 0.59375.

176. The correct answer is b) The binary fraction 0.111 is equivalent to the decimal number 0.875.

177. The correct answer is c) The binary fraction 0.0101 is equivalent to the decimal number 0.53125.

178. The correct answer is c) The binary fraction 0.10101 is equivalent to the decimal number 0.90625.

179. The correct answer is c) The binary fraction 0.0111 is equivalent to the decimal number 0.59375.

180. The correct answer is a) The binary fraction 0.11011 is equivalent to the decimal number 0.671875.

181. The correct answer is b) The decimal fraction 0.125 is equivalent to the octal fraction 0.2.

182. The correct answer is c) The decimal fraction 0.375 is equivalent to the octal fraction 0.3.

183. The correct answer is c) The decimal fraction 0.625 is equivalent to the octal fraction 0.6.

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184. The correct answer is b) The decimal fraction 0.875 is equivalent to the octal fraction 0.7.

185. The correct answer is c) The decimal fraction 0.25 is equivalent to the octal fraction 0.4.

186. The correct answer is c) The decimal fraction 0.5 is equivalent to the octal fraction 0.6.

187. The correct answer is b) The decimal fraction 0.0625 is equivalent to the octal fraction 0.05.

188. The correct answer is c) The decimal fraction 0.03125 is equivalent to the octal fraction 0.04.

189. The correct answer is b) The decimal fraction 0.15625 is equivalent to the octal fraction 0.2.

190. The correct answer is d) The decimal fraction 0.78125 is equivalent to the octal fraction 0.8.

191. The correct answer is b) The octal fraction 0.2 is equivalent to the decimal number 0.25.

192. The correct answer is b) The octal fraction 0.4 is equivalent to the decimal number 0.25.

193. The correct answer is b) The octal fraction 0.6 is equivalent to the decimal number 0.5.

194. The correct answer is c) The octal fraction 0.7 is equivalent to the decimal number 0.75.

195. The correct answer is c) The octal fraction 0.3 is equivalent to the decimal number 0.375.

196. The correct answer is c) The octal fraction 0.5 is equivalent to the decimal number 0.4375.

197. The correct answer is b) The octal fraction 0.05 is equivalent to the decimal number 0.03125.

198. The correct answer is b) The octal fraction 0.04 is equivalent to the decimal number 0.03125.

199. The correct answer is b) The octal fraction 0.2 is equivalent to the decimal number 0.25.

200. The correct answer is c) The octal fraction 0.8 is equivalent to the decimal number 0.875.

201. The correct answer is b) The octal fraction 0.2 is equivalent to the binary fraction 0.010.

202. The correct answer is b) The octal fraction 0.4 is equivalent to the binary fraction 0.010.

203. The correct answer is c) The octal fraction 0.6 is equivalent to the binary fraction 0.110.

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204. The correct answer is a) The octal fraction 0.7 is equivalent to the binary fraction 0.111.

205. The correct answer is b) The octal fraction 0.3 is equivalent to the binary fraction 0.010.

206. The correct answer is b) The octal fraction 0.5 is equivalent to the binary fraction 0.011.

207. The correct answer is c) The octal fraction 0.05 is equivalent to the binary fraction 0.001010.

208. The correct answer is c) The octal fraction 0.04 is equivalent to the binary fraction 0.001010.

209. The correct answer is b) The octal fraction 0.2 is equivalent to the binary fraction 0.010.

210. The correct answer is a) The octal fraction 0.8 is equivalent to the binary fraction 0.111.

211. The correct answer is b) The binary fraction 0.001 is equivalent to the octal fraction 0.2.

212. The correct answer is b) The binary fraction 0.010 is equivalent to the octal fraction 0.2.

213. The correct answer is d) The binary fraction 0.011 is equivalent to the octal fraction 0.6.

214. The correct answer is a) The binary fraction 0.111 is equivalent to the octal fraction 0.7.

215. The correct answer is c) The binary fraction 0.00101 is equivalent to the octal fraction 0.14.

216. The correct answer is c) The binary fraction 0.0101 is equivalent to the octal fraction 0.24.

217. The correct answer is b) The binary fraction 0.000101 is equivalent to the octal fraction 0.06.

218. The correct answer is c) The binary fraction 0.001101 is equivalent to the octal fraction 0.17.

219. The correct answer is d) The binary fraction 0.1001 is equivalent to the octal fraction 0.49.

220. The correct answer is b) The binary fraction 0.11101 is equivalent to the octal fraction 0.76.

221. The correct answer is b) The hexadecimal fraction 0.1 is equivalent to the binary fraction 0.0010.

222. The correct answer is c) The hexadecimal fraction 0.3 is equivalent to the binary fraction 0.0110.

223. The correct answer is d) The hexadecimal fraction 0.5 is

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equivalent to the binary fraction 0.1100.

224. The correct answer is d) The hexadecimal fraction 0.7 is equivalent to the binary fraction 0.1110.

225. The correct answer is b) The hexadecimal fraction 0.2 is equivalent to the binary fraction 0.0100.

226. The correct answer is a) The hexadecimal fraction 0.4 is equivalent to the binary fraction 0.0100.

227. The correct answer is c) The hexadecimal fraction 0.06 is equivalent to the binary fraction 0.011000

228. The correct answer is c) The hexadecimal fraction 0.08 is equivalent to the binary fraction 0.010000.

229. The correct answer is c) The hexadecimal fraction 0.09 is equivalent to the binary fraction 0.010000.

230. The correct answer is c) The hexadecimal fraction 0.3A is equivalent to the binary fraction 0.01111010.

231. The correct answer is c) The binary addition of 1101 and 1011 yields 11110.

232. The correct answer is b) The binary addition of 101010 and 110011 yields 1111101.

233. The correct answer is c) The binary addition of 1111 and 1111 yields 10000.

234. The correct answer is c) The binary addition of 1010101 and 1101010 yields 10011111.

235. The correct answer is a) The binary addition of 10010 and 111001 yields 101011.

236. The correct answer is d) The binary addition of 11101 and 1001 yields 111010.

237. The correct answer is a) The binary addition of 1111111 and 1 yields 10000000.

238. The correct answer is c) The binary addition of 1001 and 10 yields 1101.

239. The correct answer is b) The binary addition of 11001100 and 1010101 yields 111011101.

240. The correct answer is d) The binary addition of 111 and 10101 yields 110101.

241. The correct answer is c) The binary subtraction of 1101 from 10101 yields 10110.

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242. The correct answer is c) The binary subtraction of 101010 from 110101 yields 100011.

243. The correct answer is b) The binary subtraction of 11111 from 100000 yields 10001.

244. The correct answer is b) The binary subtraction of 10101 from 110010 yields 100110.

245. The correct answer is a) The binary subtraction of 10010 from 111001 yields 100011.

246. The correct answer is c) The binary subtraction of 1001 from 11101 yields 1101.

247. The correct answer is b) The binary subtraction of 10000000 from 11111111 yields 111.

248. The correct answer is d) The binary subtraction of 10 from 1001 yields 1000.

249. The correct answer is d) The binary subtraction of 1010101 from 11001100 yields 1011101.

250. The correct answer is a) The binary subtraction of 10101 from 111111 yields 100010.

251. The correct answer is b) The 1's complement of 11001 is 00110. Each bit is inverted (0 becomes 1, and 1 becomes 0).

252. The correct answer is b) The 1's complement of 1010101 is 0101010.

253. The correct answer is b) The 1's complement of 111000 is 000111.

254. The correct answer is c) The 1's complement of 1001011 is 0110101.

255. The correct answer is d) The 1's complement of 11111 is 11110.

256. The correct answer is c) The 1's complement of 1011 is 0101.

257. The correct answer is d) The 1's complement of 11111111 is 11111110.

258. The correct answer is b) The 1's complement of 110 is 001.

259. The correct answer is b) The 1's complement of 1000101 is 0111010.

260. The correct answer is c) The 1's complement of 1110000 is 0001110.

In 1's complement, each bit of the binary number is inverted or flipped (0 becomes 1, and 1 becomes 0) to obtain its 1's complement representation.

261. d) The 2's complement of 11001 is 11110. To obtain the 2's complement, the bits are inverted (0 becomes 1, and 1 becomes 0),

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and then 1 is added to the least significant bit.

262. b) The 2's complement of 1010101 is 0101010.

263. c) The 2's complement of 111000 is 000110.

264. b) The 2's complement of 1001011 is 0110100.

265. d) The 2's complement of 11111 is 11111 (no change).

266. d) The 2's complement of 1011 is 1101.

267. b) The 2's complement of 11111111 is 00000000.

268. b) The 2's complement of 110 is 001.

269. c) The 2's complement of 1000101 is 0111011.

270. c) The 2's complement of 1110000 is 0001110.

In 2's complement, the positive numbers remain the same, while the negative numbers are obtained by taking the 1's complement and adding 1 to the least significant bit.

271. The correct answer is b) To subtract 1 from 7 using 1's complement in base 10, we first find the 1's complement of 1, which is 8. Then we add 8 to 7, resulting in 15. However, since there is a carry from

the leftmost digit, we add 1 to the result, giving us 16. Finally, we discard the leftmost carry bit, resulting in 6.

272. The correct answer is a)

273. The correct answer is a) To subtract 5 from 9 using 1's complement in base 10, we first represent the numbers in 1's complement form. 5 in 1's complement is -4, and 9 in 1's complement is -9.

Next, we perform the subtraction using regular addition. $(-9) + (-4) = -13$.

Since the result is negative, we take the 1's complement of -13, which is 12. Finally, we add a negative sign to the result to indicate that it is negative.

274. The correct answer is c) 1110.

To subtract 7_{10} from 1_{10} using 1's complement, we first convert the numbers to their 1's complement representation.

1_{10} is represented as 0001 in 1's complement.

7_{10} is represented as 1000 in 1's complement.

Next, we perform the subtraction using regular addition. $0001 + 1000 = 1001$.